BANKHAUL · PROBLEM STATEMENT

BankHaul problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Kit 1.8: users-and-needs and stakeholders tables (Tables 2 and 3), the sources of the user requirements in BKH-URS-001
0.2	2026-10-03	Amish Chadha	TRL 2 update; first co-design candidate, safety note, open questions routed to the decisions register
0.1	2026-09-30	Amish Chadha	Initial scaffold

BankHaul problem statement

Small-scale fishers often work gillnets and traps close to shore, in the shallows or from canoes. In crocodile water, fishing is the activity during which most recorded attacks happen.

The problem

Attack studies across Africa and Asia point to the same activity. At Kariba, fishing accounted for 58 % of sampled attacks, ahead of domestic chores at 27 % (Oryx). In northern Mozambique victims were collecting water, bathing, wading across rivers, falling from canoes and fishing with gill nets, rod and line, or nets and traps (Oryx, 2010). In Timor-Leste, fishing and crabbing account for about 80 % of reported attacks (ABC News, 2025).

The main engineered answer so far is the crocodile exclusion enclosure, a fenced area at the water's edge. Enclosures work well for bathing, collecting water and washing, but are far less useful for fishers (CrocAttack). The Uganda Wildlife Authority built enclosures on Yuwe Island in Lake Victoria in 2019 after about 10 deaths, mainly to protect water collection (PML Daily, 2019). Commercial setnet fishers already use running lines, a pulley system from shore to deeper water, to moor skiffs (Northwest Setnetters), but that practice assumes boats and has not been packaged as a no-wade method for subsistence fishers.

Users and context

User	Need	Context
Subsistence and small-scale fishers	Set, check and haul nets from the bank at night and at dawn	Lake shores, river banks, estuaries and mangrove edges
Fisher associations and village committees	A shared set of stations at known hotspots	Landing sites and fishing camps
Wildlife authorities	A mitigation measure they can promote alongside patrols and closures	Human-wildlife conflict programmes
NGOs and livelihood programmes	A low-cost, locally made kit	Households in conflict hotspots

Users and needs

Table 2. Users and needs (the source of the user requirements in BKH-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Subsistence and small-scale fisher	Set and haul their own gillnet without wading; enough catch for the household; a pull one person can manage	Sets a 20 to 30 m gillnet at dusk from the bank, hauls at dawn, clears fish on the bank	Wading at night and dawn is when attacks happen; no money for a boat; nets snag and tear
Fisher association or village committee	A shared, durable station at a known hotspot that members can maintain	Builds and keeps stations at a landing site; agrees who uses them and when	Cash for parts; theft of metal parts; siting on banks that move with the season
Station builder (local welder, carpenter, rope worker)	Parts that can be cut, drilled, welded, sawn and spliced with local tools	Builds one station from the drawings and build plan in a few days	Unknown steel grades; no rated rigging nearby; no test rig
Boat crew placing the far stake	A stake two people can lift and drive from a boat in one trip	Places or moves the far stake once a season at low water or from a moored boat	The most exposed job in the system; soft bed; no lifting gear

Stakeholders

Table 3. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Wildlife authority (for example Zimbabwe Parks and Wildlife Management Authority, Uganda Wildlife Authority)	Human-wildlife conflict mitigation; crocodile management; safety advice at the site	Agree siting, safety briefing and any permit (PR-016)
Fisheries authority	Fishing licences, gear and mesh rules, closed zones and seasons	Confirm that loop-set gillnets are allowed where stations are sited (BKH-REG-001)
Human-wildlife conflict researchers (for example the Lake Kariba group of the Oryx study)	Attack data by place, time and activity; trial design	Co-design partner candidate (BKH-DDR-001, item 10); site selection
NGOs and livelihood programmes	Funding and distribution of low-cost kits	Costing of the local-materials station (PR-008)
Other water users (boats, other fishers)	Navigation past a floating line	Marking of the loop and the far stake

Operating environment

- Fresh or brackish water, still to slow flowing; banks from firm to soft mud and sand.
- Working span from bank anchor to offshore pulley of 10 to 50 m (33 to 164 ft) (target).
- Water depth at the offshore pulley up to about 3 m (10 ft) for a stake, deeper with a float (estimate).
- Seasonal level change of the shoreline; banks move as water rises and falls.
- Night and dawn use, sun, UV and floating debris; hippos and crocodiles present.

Constraints

- Prototype budget ceiling USD 1,800 for all prototypes and test rigs.
- No powered hauler; hand pull only (IP screen design-around).
- Parts cost target below USD 80 per station (target), using commodity rope, pulleys and anchors.
- Installable without anyone standing in crocodile water; the offshore point is placed from a boat, at low water, or by a trained team with protection.
- Open hardware under CERN-OHL-S-2.0.

Out of scope

- Crocodile deterrence, capture or culling.
- Protection for bathing and water collection, already covered by exclusion enclosures.
- Motorised line haulers or winches.
- Changes to fishing regulations or zoning.

Prior work

Prior work	What it does	Gap for these users	Source
Crocodile exclusion enclosures (CEEs)	Metal or wire fenced areas at the water's edge for bathing, washing and collecting water	Far less useful for fishers, who need to reach water beyond the fence	link
Yuwe Island enclosures, Uganda Wildlife Authority (2019)	Crocodile exclusion enclosure for residents who depend on Lake Victoria for water	Protects a fixed water point, not fishing	link
Setnet running lines	A pulley line from shore to deeper water used by commercial setnetters to moor skiffs	Assumes boats and crew; not documented as a no-wade method for subsistence nets	link
EP1429601B1, device for hauling a fishing line (expired 2022)	A hydraulic motorised pulley and capstan line hauler	Powered and boat-mounted; BankHaul stays manual and bank-based	link

Co-design

First candidate to approach (not agreed): the Lake Kariba research group behind the Oryx study (Matanzima, Marowa and Nhiwatiwa), together with an inshore fishing cooperative on the Zimbabwean shore of the lake, because the researchers hold attack locations and times and can site a trial station where attacks happen (BKH-DDR-001, item 10).

Checklist for the co-design work:

- [ ] Confirm the net types, lengths and depths used at the trial site.
- [ ] Map the bank, bed and water level over a season at two candidate stations.
- [ ] Agree who places and inspects the far stake, and from which boat.

- [ ] Agree the safety briefing with the local wildlife authority.

The original brief: a fisher association in a documented attack hotspot, working with the wildlife authority or a human-wildlife conflict researcher who already holds local attack data, so the station is sited where attacks happen and tested with the nets people actually use.

Safety

> **Safety:** BankHaul is a rope system worked beside crocodile water. It reduces time in the water; it does not protect anyone from a crocodile that leaves the water. Placing the far stake is the most dangerous step and is done from a boat or at low water, never by wading. A loaded rope can snap back, and a net seized by a large animal can pull hard on the loop: the design limits that pull with a weak link, keeps the fisher at least 5 m from the water and never ties the loop to anyone. BankHaul is published as an open engineering reference, not certified equipment.

Open questions

The questions below are answered, or routed to the design decisions register (`docs/06-design-decisions.md`), at TRL 3.

- How can the offshore stake or float be placed safely where no boat is available?
- Does a moving loop disturb fish or crocodiles more than a static net?
- Which net types (gillnet, seine, traps) suit the loop, and which do not?
- Will fishers use shared stations, or does each household need its own?
- How does anchor pull-out behave in soft, saturated beds over a season?